

give me opps interview question s

Since you're preparing for placements, these are the **most frequently asked OOP (Object-Oriented Programming) interview questions** in C++. If you can answer these confidently with examples, you'll be prepared for most interviews.

Basic OOP Questions (Very Frequently Asked)

1. What is Object-Oriented Programming (OOP)?



2. What are the four pillars of OOP?

- Encapsulation
- Abstraction
- Inheritance
- Polymorphism

★ Almost every interviewer asks this.

3. What is a class?

4. What is an object?

5. Difference between class and object?

6. Why do we need OOP instead of procedural programming?

Encapsulation

7. What is encapsulation?

8. How do getters and setters help achieve encapsulation?

9. Why should data members be private?

10. Can encapsulation exist without getters and setters?

Abstraction

11. What is abstraction?

12. Difference between abstraction and encapsulation?

★ Very common.

13. How is abstraction achieved in C++?

14. Give a real-life example of abstraction.

Constructors

15. What is a constructor?

16. Types of constructors?

- Default
- Parameterized
- Copy

17. What is a copy constructor?

18. Why is a copy constructor needed?

19. Difference between copy constructor and assignment operator?

☆☆ Frequently asked.

20. Can constructors be virtual?

Answer: No.



21. Can constructors be overloaded?

Answer: Yes.

22. Can constructors return values?

Answer: No.

23. Can a constructor be private?

Answer: Yes.

Destructor

24. What is a destructor?

25. Why do we need destructors?

26. Can destructors be overloaded?

Answer: No

27. Can destructors be virtual?

Answer: **Yes**

Very important.

Inheritance

28. What is inheritance?

29. Types of inheritance?

- Single
- Multiple
- Multilevel
- Hierarchical
- Hybrid

30. Difference between public, protected and private inheritance?

31. What is the diamond problem?

32. How is the diamond problem solved?

Answer:

Virtual inheritance.

33. Difference between protected and private access specifiers?

34. Can constructors be inherited?



35. Order of constructor and destructor calls in inheritance?

Interview favourite.

Polymorphism

36. What is polymorphism?

37. Types of polymorphism?

- Compile-time
- Run-time



38. What is function overloading?

39. What is operator overloading?

40. What is function overriding?

41. Difference between overloading and overriding?

☆☆ Very common.

42. What is virtual function?

43. Why do we use virtual functions?

44. What is dynamic binding?

45. What is early binding?

46. What is late binding?

47. Can static functions be virtual?

Answer: No

48. What is pure virtual function?

49. What is an abstract class?

50. Can we instantiate an abstract class?

Answer: No

Friend

51. What is a friend function?

52. Why use friend functions?

53. What is a friend class?

Static

54. What is a static data member?

55. What is a static member function?



56. Why can't static member functions access non-static members?

● This Pointer

57. What is the `this` pointer?

58. Why do we use the `this` pointer?

59. Can we return `*this` ?



● Memory

60. What is stack memory?

61. What is heap memory?

62. Difference between stack and heap?

Very common.

63. What does `new` do?

64. What does `delete` do?

65. Difference between `malloc()` and `new` ?

66. Difference between `free()` and `delete` ?

Pointers & Objects

67. Object created on stack vs heap?

68. What is a pointer to an object?

69. Why use `->` instead of `.` ?

70. What happens when we delete an object?



Access Specifiers

71. Difference between

- public
- private
- protected

72. Default access specifier of a class?

Answer:

Private.

73. Default access specifier of a struct?

Answer:

Public.

Virtual Keyword

74. Why is the destructor made virtual?

Very important.

75. What happens if the base class destructor isn't virtual?

76. What is a virtual base class?

Miscellaneous

77. Can we have multiple constructors?

78. Can we overload destructors?

79. Difference between object composition and inheritance?

80. What is object slicing?

Frequently asked by product-based companies.

81. What is the Rule of Three?

82. What is the Rule of Five?

83. What is the Rule of Zero?

84. What is an interface in C++?

85. What is multiple inheritance?



86. Why doesn't Java support multiple inheritance using classes?

(Asked for OOP concepts.)

87. What is the difference between `struct` and `class` ?

88. What is a namespace?

89. What is an inline function?

≡ ChatGPT ▾



90. What are const member functions?

Coding Questions

Interviewers may ask you to write code for:

- Write a class with constructor and destructor.
- Implement encapsulation using getters and setters.
- Create a copy constructor.
- Demonstrate inheritance.
- Demonstrate function overloading.
- Demonstrate function overriding.
- Implement runtime polymorphism using virtual functions.
- Create an abstract class.
- Implement a singleton class (uses a private constructor).
- Explain constructor/destructor order using code.
- Explain object slicing with code.
- Show shallow copy vs deep copy.
- Implement operator overloading (e.g., `+`, `==`, `<<`).
- Explain stack vs heap object creation using code.

Must-Prepare Questions (Top 20)

If you have limited time, focus on these:

1. Four pillars of OOP

2. Encapsulation
3. Abstraction vs Encapsulation
4. Inheritance
5. Types of inheritance
6. Polymorphism
7. Compile-time vs Runtime polymorphism
8. Function overloading vs Overriding
9. Virtual functions
10. Pure virtual functions
11. Abstract class
12. Constructor types
13. Copy constructor
14. Destructor
15. Virtual destructor
16. Stack vs Heap
17. new vs malloc
18. this pointer
19. Object slicing
20. Shallow copy vs Deep copy



Since you've recently been learning **copy constructors, encapsulation, and pointers**, the next topics I'd recommend are:

1. Constructors & Destructors
2. Inheritance
3. Polymorphism (especially virtual functions)
4. Shallow Copy vs Deep Copy
5. Virtual Destructor
6. Object Slicing
7. Rule of Three/Five

These topics come up repeatedly in C++ placement interviews at companies ranging from service-based firms to product-based companies like Atlassian, Adobe, and Microsoft.



